Business Dynamics Statistics Executive Dashboard

1. Project Overview

This project presents an interactive executive dashboard built with Microsoft Power BI, utilizing the publicly available Business Dynamics Statistics (BDS) datasets published by the U.S. Census Bureau. The dashboard is designed to support economic analysis, business policy evaluation, and strategic decision-making by visualizing key metrics related to business formation, employment changes, establishment entries and exits, and firm dynamics across multiple dimensions – including U.S. states, industry sectors, firm size categories, and firm age groups.

The primary purpose of this project is to demonstrate a complete data analytics workflow: from data acquisition and transformation, through star-schema data modelling, to the development of an interactive dashboard and its deployment to Power BI Service with a public shareable link. The final output serves as a tool for exploring how economic activity evolves over time, which sectors drive job creation, and how small versus large firms contribute to net employment growth.

2. Data Source and Description

The datasets used originate from the Business Dynamics Statistics (BDS) program, which provides annual measures of business dynamics – such as job creation and destruction, firm births and deaths, and establishment expansions and contractions. The BDS data are derived from the Longitudinal Business Database (LBD) and are widely used by economists, policymakers, and researchers.

All files contain additional metrics such as establishment entry and exit rates, job creation from births and continuers, job destruction from deaths and continuers, net job creation, and reallocation rates. The data span multiple years (depending on the version used) and cover the entire U.S. private non-farm economy.

3. Relevance to Real-World Economic Analysis

The BDS data are uniquely valuable for economic analysis because they track the actual dynamics of businesses over time, not just static counts. The key economic questions that this dashboard helps address include:

- **Job creation and destruction**: Which sectors or states generate the most net new jobs? Are jobs being created by new firms (births) or by existing firms expanding?
- **Business churn**: High rates of establishment entry and exit can indicate a dynamic, competitive economy, but may also signal instability. The dashboard allows comparison of entry and exit rates across sectors.
- **Firm size and age dynamics**: Do small firms create more jobs relative to their share of employment? Do older firms become more stable but less innovative? The dashboard provides visual answers.
- **Regional economic performance**: By filtering by state, the dashboard helps identify regional disparities and can inform targeted economic development policies.

In a policy or business context, such insights guide decisions on where to allocate investment, which industries to support, and how to measure the effectiveness of entrepreneurship programs.

4. Methodology and Data Modelling

4.1 Data Preparation in Power Query

All CSV files were loaded into Power BI Desktop. The following transformation steps were applied:

- Removal of duplicate rows and handling of missing values (where necessary).
- Renaming of tables to reflect their grain (e.g., Fact_StateSector, Fact_State).
- Extraction of distinct dimension values: from the main fact table (state_by_sector), unique years, state codes (st), and sector codes were extracted to create separate dimension tables DimDate, DimState, and DimSector.
- Similarly, from the firm_size and firm_age tables, distinct fsize and fage values were extracted to create DimFirmSize and DimFirmAge.

4.2 Star Schema Design

A star schema was implemented to ensure efficient querying, ease of maintenance, and accurate cross-filtering. The model consists of:

- **One fact table** (Fact_StateSector) that holds the most granular data at the intersection of state, sector and year. Additional fact tables (Fact_State, Fact_Sector, Fact_FirmSize, Fact_FirmAge) were retained for specific analyses but are linked through shared dimensions.
- **Five dimension tables**: DimDate, DimState, DimSector, DimFirmSize, DimFirmAge.

Relationships are all one-to-many from dimension to fact, with single direction cross-filtering. This avoids ambiguous many-to-many relationships and ensures that slicers on years, states or sectors correctly filter all relevant fact tables.

4.3 DAX Measures

Several measures were created to support Key Performance Indicators (KPIs):

- Total Firms = SUM(Fact_StateSector[firms])
- Total Employment = SUM(Fact_StateSector[emp])
- Net Job Creation = SUM(Fact_StateSector[net_job_creation])
- Job Creation Rate = AVERAGE(Fact_StateSector[job_creation_rate])
- Job Destruction Rate = AVERAGE(Fact_StateSector[job_destruction_rate])
- Estabs Entry Rate = AVERAGE(Fact_StateSector[estabs_entry_rate])
- Estabs Exit Rate = AVERAGE(Fact_StateSector[estabs_exit_rate])

5. Dashboard Features

The final dashboard comprises three interactive pages, each focusing on a different analytical perspective.

5.1 Page 1 – Executive Overview

This page provides a high-level summary of the entire economy:

- **KPI cards** displaying total firms, total employment, net job creation, job creation rate, and job destruction rate.
- **Line chart** showing the trend of job creation rate versus job destruction rate over the available years.
- **Filled map** of the United States, with colour intensity representing total employment by state.
- **Donut chart** breaking down total firms by industry sector.
- **Slicers** for year, sector and state, placed at the top for easy filtering.

All visuals are cross-filtered by the slicers, enabling the user to, for example, examine only the manufacturing sector in California over the last decade.

5.2 Page 2 – Sector and State Deep Dive

This page is designed for comparative analysis:

- **Bar chart** ranking sectors by total employment.
- **Line chart** showing employment trends over time, with lines for different sectors.
- **Matrix heat map** with states as rows and sectors as columns, colour-coded by total employment to quickly spot strong and weak combinations.
- **Scatter plot** of job creation rate against establishment exit rate, with each point representing a sector; this reveals which sectors have high churn.
- **State performance table** listing states with their total firms, employment and net job creation.

5.3 Page 3 – Firm Demographics

This page focuses on the structure of firms by size and age:

- **Bar chart** of total firms by firm size category (e.g., 1-4 employees, 5-9, etc.).
- **Bar chart** of total firms by firm age category (e.g., 0-1 years, 2-3 years, etc.).
- **Line chart** of job creation rate over time, with separate lines for different firm size categories – illustrating whether small or large firms are leading job growth.
- **Waterfall chart** showing net job creation contributions by firm size category (increases and decreases).

Slicers for year, firm size and firm age are synchronized across this page.

5.4 Interactivity and Navigation

Slicers are synchronized across all three pages using the Sync Slicers pane in Power BI. This means that a user selecting year 2015 on Page 1 will automatically see the same year applied on Pages 2 and 3.

Clicking on any visual (e.g., a sector in the donut chart) cross-filters other visuals on the same page, allowing intuitive data exploration.

6. Key Insights Obtained from the Dashboard

While the exact numbers depend on the specific time period covered by the dataset, the dashboard typically reveals the following patterns (illustrative, based on typical BDS data):

- Job creation rates are generally higher in the services and construction sectors compared to manufacturing, which tends to have lower but more stable job creation.
- Establishment entry rates are highest among very small firms (1-4 employees) and among young firms (0-2 years old). However, their survival rates vary significantly by sector.
- Net job creation is positive in most years, but during economic downturns, job destruction rates exceed creation rates, leading to negative net job creation. The dashboard allows a quick visual comparison of those periods.
- Some states (e.g., Texas, Florida) consistently show higher net job creation than others, driven by a combination of firm births and expansions. The heat map matrix makes it easy to see which sectors are responsible for that advantage.
- Older firms (10+ years) contribute less to job creation as a percentage of their employment base than young firms, but they provide the majority of stable employment.

7. Conclusion

This Power BI project successfully demonstrates how raw public economic data can be transformed into an actionable decision-support tool. By applying a star schema model, creating intuitive visualizations, and deploying to a public web environment, the dashboard makes Business Dynamics Statistics accessible to a wide audience. The insights derived – such as the role of young small firms in job creation and the regional variation in economic dynamism – have direct relevance to economic policy, business strategy, and academic research. The documented methodology also serves as a reusable template for similar data analytics projects.